

Theme: Cyber Safety

TrustLayer

One-Line Solution:

TrustLayer analyzes digital media across multiple verification layers to detect synthetic content & provenance inconsistencies.

Key Focus Areas

- Synthetic Content & Deepfake Detection
- Digital Media Provenance & Authenticity Verification
- Multi-Layered Security & Cyber Safety Standards



TEAM BIRYANI

Team Members

Member 1: Md Kasif Uddin

Member 2: Raj Vardhan Pal

Member 3: Avneet Kaur

Member 4: Vivek Singh

PROBLEM

What is the problem?

Fake & AI-generated content

Images, videos, screenshots, and social media posts are spreading rapidly across the internet.

Verification difficulty

It is incredibly hard for the average person to tell what is real and what is fabricated.

Lack of tools

Most people have no easy, accessible way to check the authenticity of the media they consume daily.



Who faces it?

- **Social media users:** Everyday consumers.
- **Students:** Researching academic topics.
- **Journalists:** Verifying news sources.
- **General public:** Online readers/sharers.

Why is it important?

- **Scam vulnerability:** False news & financial scams.
- **Reputation damage:** Fake edited screenshots.
- **Erosion of trust:** Harder to trust digital media.

PROPOSED SOLUTION

TrustLayer is a web tool where a person adds something found online (text, image, video, screenshot, link). It runs checks and gives a trust score with a short reason.

How It Solves The Problem

Centralized checking:

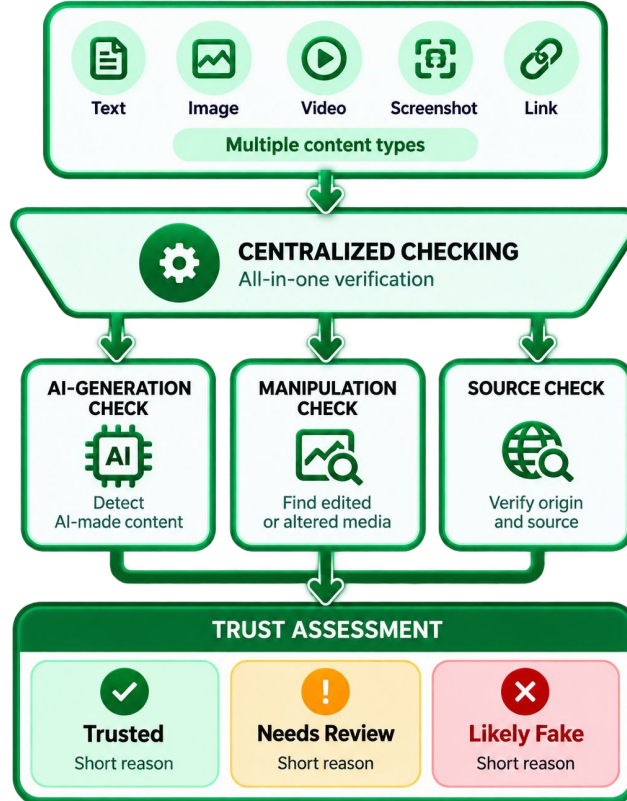
Checks many types of content in one single place.

Clear scoring:

Gives a green, yellow, or red score that anyone understands immediately.

Contextual reasoning:

Provides a clear reason so the user knows exactly why the score was given.



What Makes It Different

Comprehensive coverage:

Covers text, images, videos, screenshots, and links together in one platform.

User-friendly:

No technical knowledge or background is needed to use the tool.

Actionable insights:

The score comes with a clear reason, not just a simple yes or no.



CHANDIGARH UNIVERSITY

Discover. Learn. Empower.

KEY FEATURES



Checking The Content

AI-generated content detection

Finds text, images, or video made by artificial intelligence engines.

Manipulated media detection

Spots edited, altered, or changed photos and videos that attempt to deceive.

Information source checking

Looks deeply at original sources to establish baseline credibility.



Catching Fakes & Results

Fake screenshot identification

Flags screenshots, chat logs, and posts that were entirely made up or edited.

Simple trust score

Condenses technical analysis into a clear color-coded score with concise explanation.



CHANDIGARH UNIVERSITY

Discover. Learn. Empower.

TECHNOLOGY & IMPLEMENTATION



Frontend

Technology: NextJs, FastAPI, and C2PA

Purpose: The responsive page where the user easily adds content and views their final trust score.

Backend

Technology: Python and Node.js

Purpose: Receives the submitted content, securely runs the verification checks, and sends back the processed result.

Database

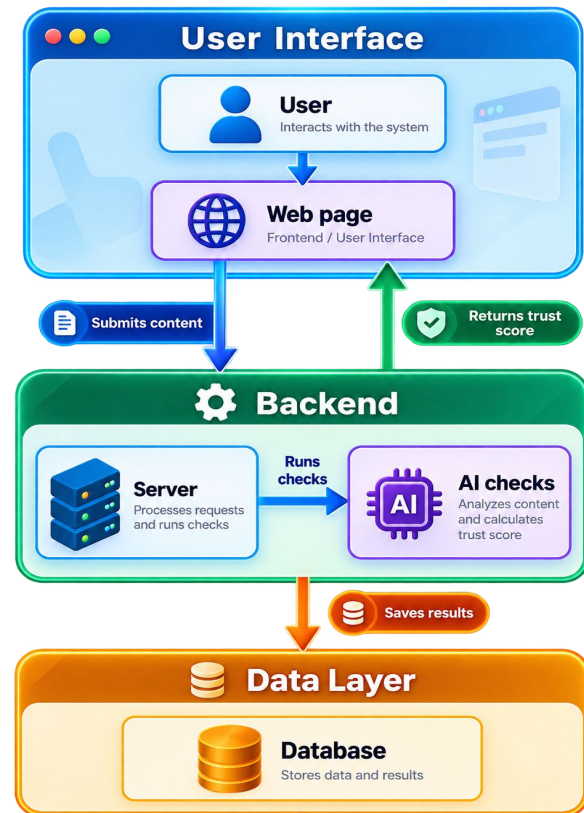
Technology: Supabase and MySQL

Purpose: Safely saves historical results, user queries, and other essential application data.

AI

Technology: Python and LLM API

Purpose: Runs the complex AI detection checks and automatically writes the short, readable reason for the score.



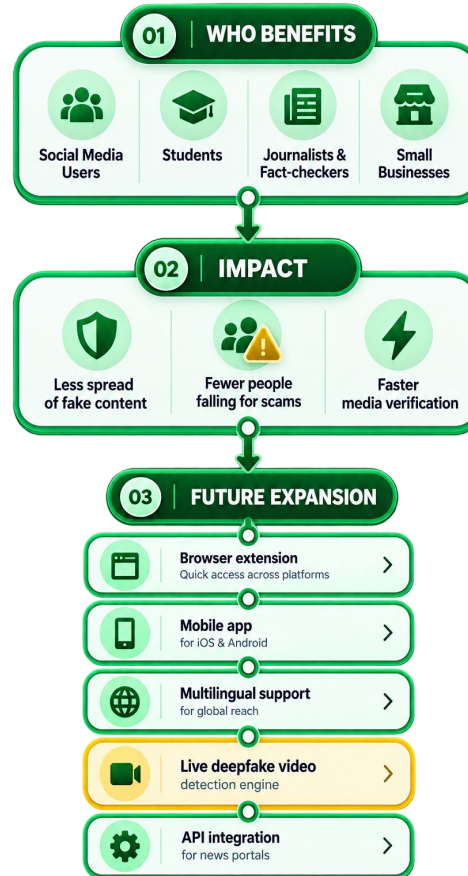
IMPACT & FUTURE SCOPE

Who Benefits?

- **Social media users:** Check posts before sharing
- **Students:** Verify research sources
- **Journalists & Fact-checkers:** Fast primary check
- **Small businesses:** Catch fake screenshots

What Impact Will It Create?

- **Less spread** of fake content
- **Fewer people** falling for scams
- **Faster** media verification



How Can It Be Expanded?

- **Browser extension** for quick access
- **Mobile app** for iOS & Android
- **Multilingual support** for global reach
- **Live deepfake video** detection engine
- **API integration** for news portals

**"Know what's real
before you trust it."**